

Esha Vijay

Data Analyst/ Data Scientist

CONTACT INFORMATION:

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eshavijay55@gmail.com

SKILLS:

Programming Languages: Python, SQL, R, Java

Database Technologies: MySQL, MongoDB, PostgreSQL

Version Control: Git

Visualization Software: PowerBI, Microsoft Excel, Tableau

Machine Learning: TensorFlow, PyTorch, Scikit-Learn

Strong problem-solving and analytical skills
Excellent communication and leadership abilities

LANGUAGES:

English
Tamil
Hindi
Kannada

REFERENCES:

Available upon request.

PROFESSIONAL SUMMARY:

Dynamic data analyst with a robust foundation in computer science and hands-on experience in IoT project development. Leveraging expertise in Python programming and SQL proficiency, proficiency enables the transformation of complex datasets into actionable insights. Demonstrated ability in developing unit testing frameworks and cloud deployment enhances the commitment to delivering high-quality solutions. Eager to contribute analytical skills and innovative problem-solving approaches to drive impactful results in the data science field. Passionate about harnessing data to inform strategic decision-making and optimize performance.

WORK EXPERIENCE:

IOT INTERN

Engaged in the end-to-end development and deployment of IoT projects, emphasizing device management and cloud solutions. Key contributions and responsibilities included:

IoT Device Management: Designed and implemented frameworks for efficient IoT device management, enabling seamless communication and control of connected devices.

Unit Test Framework Development: Developed a comprehensive unit testing framework tailored for IoT applications, ensuring reliability, scalability, and adherence to quality standards.

Cloud Deployment: Successfully deployed IoT solutions to cloud platforms, optimizing performance and facilitating remote accessibility for real-time data monitoring and management. Documentation and Strategy Development: Created detailed technical documentation, strategies, and workflows to support ongoing IoT projects.

The 14-week internship involved collaboration with cross-functional teams, contributing to innovative solutions in IoT, cloud integration, and testing frameworks while maintaining high professional and technical standards.

EDUCATION:

Bachelors of Technology, Computer Science

Jain University

2018 - 2022

Masters of Science, Computer Science

University of Massachusetts Amherst

September 2023 - December 2025

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COURSES TAKEN:

- Machine Learning
- Responsible AI
- Theory and Practice of Software Engineering
- Intelligent Visual Computing
- Methods of Applied Statistics
- Data Management for Business Leaders
- Introduction to Computer & Network Security
- Decarbonization and Data Science
- Optimization in Data Science
- Data Visualization
- Artificial Intelligence

PROJECTS:

Patient Tracker System:

Collaborated on the development of a secure, user-friendly patient management system designed to streamline healthcare operations. Key contributions included implementing real-time notifications, user authentication, and data visualization features, ensuring HIPAA-compliant encryption and secure data handling. Designed architecture diagrams, developed UI mockups, and structured a relational database to optimize system performance. Conducted functional and non-functional requirement evaluations, achieving robust performance under simulated high-traffic conditions.

Titanic Prediction using Machine Learning:

Developed and evaluated machine learning models to predict passenger survival on the Titanic using Kaggle's Titanic dataset. Implemented advanced data preprocessing techniques, including handling missing values, feature engineering (e.g., extracting titles, family size metrics), and one-hot encoding for categorical data. Compared the performance of Support Vector Machines (SVM), Multi-Layer Perceptron (MLP), and Random Forest models. Conducted extensive hyperparameter tuning and cross-validation to optimize accuracy, achieving insights into feature importance and model robustness. Presented findings with performance visualizations and recommendations for future model enhancements.

Fine-Tuning Open-Source Large Language Models for Domain-Specific Applications:

Conducted fine-tuning of the Mistral 7B language model using Quantized Low-Rank Adaptation (QLoRA) for specialized tasks, including historical question answering (r/AskHistorians dataset) and poll classification (Twitter dataset). Implemented advanced techniques such as 4-bit quantization and low-rank matrix updates to reduce computational overhead while maintaining performance. Developed efficient data preprocessing pipelines, including tokenization, dataset formatting, and model sharding for GPU optimization. Evaluated model performance using metrics like ROUGE scores and conducted qualitative comparisons against GPT-4. Demonstrated improved accuracy in domain-specific text generation and classification tasks while addressing challenges of overfitting and model scalability.

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CERTIFICATION COURSES:

100 Days of Code: The Complete Python Pro Bootcamp

- UDEMY

2025 Python Data Analysis and Visualization Masterclass

- UDEMY

R Programming A-Z: R For Data Science With Real Exercises!

- UDEMY.

Microsoft Power BI Desktop for Business Intelligence

- UDEMY

Microsoft Excel: Data Visualization w/ Excel Charts & Graphs

- UDEMY

PROJECTS:

Advanced Excel Data Analysis for Financial and Sales Projection:

Executed comprehensive data analysis and financial modeling for DIG Technology using Microsoft Excel. Utilized advanced Excel tools, including Pivot Tables, Solver, and Scenario Manager, to optimize production costs, analyze sales trends, and evaluate strategic scenarios. Key contributions included designing and implementing formulas for multi-worksheet data consolidation, creating dynamic performance dashboards with slicers and charts, and running Solver to minimize production costs under predefined constraints. Enhanced decision-making through what-if analyses, data visualizations, and scenario summaries, providing actionable insights into sales performance, production optimization and revenue forecasting.

Visualizing Economic Power: A Comparative Analysis of Russia vs USA GDP:

For my final project in COMPSCI 574: Data Visualization at UMass Amherst, I developed an interactive D3.js dashboard comparing the GDP of Russia and the United States from 1990 to the present. The visualization allowed users to toggle between total GDP, GDP per capita, and global GDP share, incorporating tooltips and smooth transitions for enhanced interactivity. I focused on applying core principles of visual storytelling and data encoding to reveal economic trends and highlight the impact of key geopolitical events. The project emphasized usability, accessibility, and clarity in presenting complex temporal and comparative data.